7.

Scientists make 'new fuel' discovery

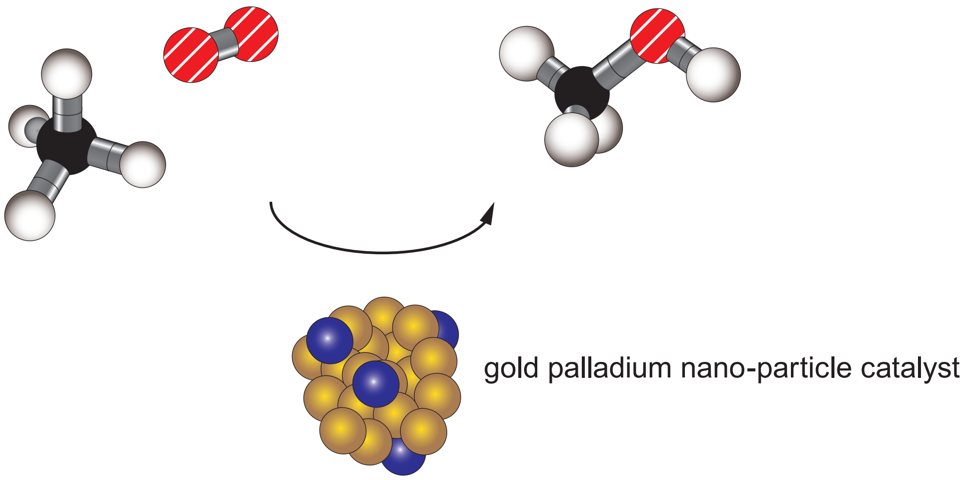
Methanol is vital in manufacturing a wide range of fuels and chemicals. Methanol is produced from methane found in natural gas. At the moment methane is condensed into liquid natural gas at the site where it is extracted and transported in pressurised containers.

Traditionally, methanol is created by converting methane into hydrogen and carbon monoxide molecules at high temperature, then rearranging the atoms in a different order in a second highly pressurised process. The current two-stage process is very energy intensive, as it burns a lot of fossil fuel to achieve high temperatures.

Stage 1 $2CH_4 + O_2 \rightleftharpoons 2CO + 4H_2$

Stage 2 $CO + 2H_2 \rightleftharpoons CH_3OH$

Scientists have discovered a new way of creating greener and cheaper methanol from methane using gold palladium nano-particles to initiate a one-stage chemical reaction that can be done at temperatures no higher than 50 °C. In this new process the gold palladium nano-particles act as a catalyst enabling methane and oxygen to combine, forming methanol in a single stage reaction.

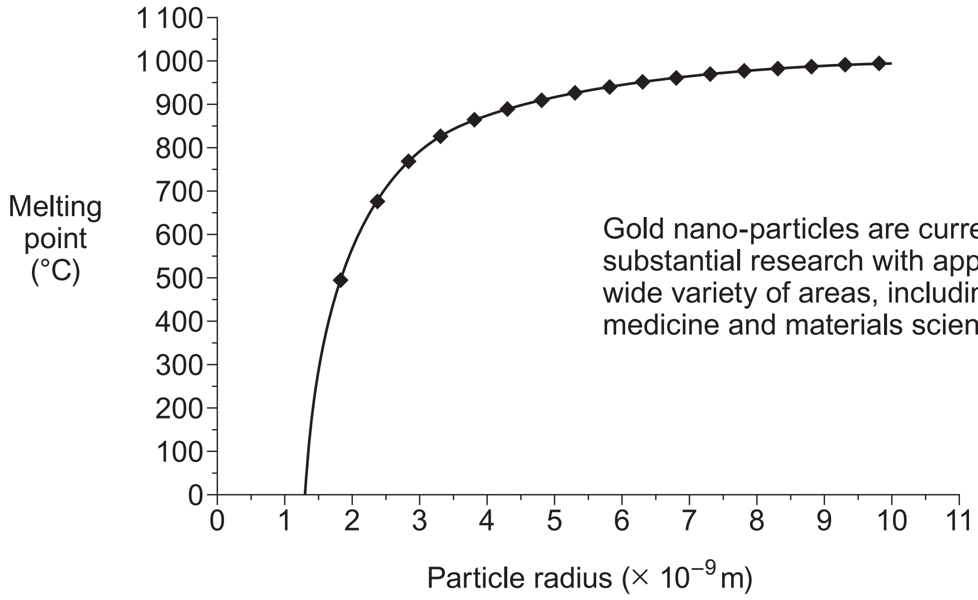


The discovery opens up the prospect of easily converting methane into methanol at the site where the methane is extracted, so that methanol can be transported as a liquid at atmospheric pressure.

Nano-particles have very different and unique properties from their bulk form.

Bulk gold	Gold nano-particles
• shiny	• found in a range of colours (100 nm purple; 20 nm red; 1 nm yellow)
• always gold colour	• never gold colour
• inert	• very good catalysts
• conducts electricity	• semi-conductors
• melting point 1 064 °C	• range of melting points

Examiner only



Gold nano-particles are currently subject to substantial research with applications in a wide variety of areas, including electronics, medicine and materials science

(a) Tick (✓) the box next to the **two** statements which support the opinion that the new method is more **environmentally friendly**. [2]

It is cheaper than the traditional method	☐
It uses less energy	☐
It reduces carbon dioxide emissions	☐
It uses gold nano-particles	☐
It uses more fuel	☐

(b) Complete the balanced equation for the **new** method of converting methane to methanol. [2]

$2CH_4$ + ... ⟶ ☐ ....

Examiner only

(c) Tick (✓) the box next to the **true** statement. [1]

The melting points of gold nano-particles and bulk gold are the same ☐

Gold nano-particles have a fixed melting point value ☐

Smaller gold nano-particles have higher melting points than larger gold nano-particles ☐

The melting point of gold nano-particles depends on their size ☐

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